TEACHER KEY

1

Measurement Evidence and SI Records

Practice answers and repair moves

L1-1. Classify 'The solution is pale blue' and 'The solution volume is 18.5 mL' as qualitative or quantitative.

ANSWER: 'Pale blue' is qualitative. '18.5 mL' is quantitative because it includes a measured value and unit.

L1-2. Choose the pictured tool best suited to measure 24.6 mL of water. Cite one visual feature.

ANSWER: The graduated cylinder; its narrow body and graduated scale support a more precise volume reading than a beaker.

L1-3. Repair each record: (a) .45g (b) 6.0ml (c) 5 kgs.

ANSWER: (a) 0.45 g (b) 6.0 mL (c) 5 kg.

L1-4. A student needs the mass of a dry solid. Explain why a graduated cylinder is not the correct first tool.

ANSWER: A graduated cylinder measures volume. An electronic balance directly measures mass.

L1-5. Write one complete quantitative record for a sample with a mass display of 7.250 g.

ANSWER: Example: sample mass = 7.250 g, measured with an electronic balance.

MISCONCEPTION	A number automatically makes an observation scientific and complete.
REPAIR	**Require students to underline the value once and circle the unit; if either mark is missing, the record is incomplete.**

TEACHER KEY

2 Scientific Notation and Significant Figures

Practice answers and repair moves

L2-1. Write 0.0005070 in scientific notation and state its significant figures.

ANSWER: 5.070×10^{-4}; four significant figures.

L2-2. State the number of significant figures in (a) 0.004060 (b) 700. (c) 1.0200 (d) 0.030.

ANSWER: (a) 4 (b) 3 (c) 5 (d) 2.

L2-3. Calculate 45.60 / 2.5 and round correctly.

ANSWER: 18; the calculator result 18.24 is rounded to two significant figures.

L2-4. Calculate 18.42 + 3.7 + 0.006 and round correctly.

ANSWER: 22.1; the tenths place limits the sum.

L2-5. A student rounds every intermediate step. State one risk and one repair.

ANSWER: Early rounding can accumulate error. Keep guard digits and round once after the final operation.

MISCONCEPTION	All zeros are either always significant or never significant.
REPAIR	**Have students label each zero as leading, captive, or trailing before counting significant figures.**

3

Dimensional Analysis

Practice answers and repair moves

L3-1. Convert 3.45 km to meters using a visible conversion factor.

ANSWER: 3.45 km x (1000 m / 1 km) = 3.45 x 10^3 m.

L3-2. Convert 0.0825 L to milliliters.

ANSWER: 82.5 mL.

L3-3. Convert 2.60 x 10^5 mg to grams and preserve significant figures.

ANSWER: 2.60 x 10^2 g, or 260. g.

L3-4. Convert 18.0 m/s to km/h.

ANSWER: 64.8 km/h.

L3-5. Explain why a setup that ends in h/s is incomplete when the target is m/s.

ANSWER: The distance unit was not converted or canceled correctly; dimensional consistency requires only m/s to remain.

MISCONCEPTION	Conversion means moving a decimal according to a memorized direction.
REPAIR	**Cover the numbers and inspect only the units. Reorient factors until every unwanted unit cancels.**

4 Density and Percent Error

TEACHER KEY

Practice answers and repair moves

L4-1. A 31.6 g sample occupies 25.5 mL. Calculate density.

ANSWER: 31.6 g / 25.5 mL = 1.24 g/mL.

L4-2. A liquid has density 1.24 g/mL and volume 25.5 mL. Calculate mass.

ANSWER: m = dV = 31.6 g.

L4-3. A solid has mass 14.9 g and density 2.50 g/cm^3. Calculate volume.

ANSWER: V = m/d = 5.96 cm^3.

L4-4. An experimental value is 12.4 g/mL and the accepted value is 12.0 g/mL. Calculate percent error.

ANSWER: |12.4 - 12.0| / 12.0 x 100 = 3.33%.

L4-5. Explain why percent error uses the accepted value in the denominator.

ANSWER: The accepted value provides the reference scale, so the difference is expressed relative to what is expected.

MISCONCEPTION	Density is the same as mass, or a larger sample must have a larger density.
REPAIR	**Compare two amounts of the same material and calculate m/V for each; the ratio stays constant while mass and volume change.**

5

Accuracy, Precision, and Error

Practice answers and repair moves

TEACHER KEY

L5-1. Trials are 10.01, 10.00, and 10.02 g; accepted mass is 10.01 g. Classify accuracy and precision.

ANSWER: The trials are precise and accurate: they cluster tightly and center on the accepted value.

L5-2. Trials are 8.42, 8.43, and 8.42 mL; accepted volume is 8.00 mL. Classify accuracy and precision.

ANSWER: Precise but not accurate relative to 8.00 mL.

L5-3. A student reads a meniscus from above. Name the error and give a specific repair.

ANSWER: Parallax/line-of-sight error. Move the eye to liquid level and read the bottom of the concave meniscus.

L5-4. A balance reads +0.20 g when empty. Explain why repeating trials without zeroing it does not solve the problem.

ANSWER: The zero offset is systematic; every result is shifted high by about 0.20 g. Zero or calibrate the balance first.

L5-5. Explain why percent error alone cannot establish precision.

ANSWER: Percent error compares with an accepted value; precision requires repeated trials to judge their agreement.

MISCONCEPTION	A low percent error proves the procedure is precise.
REPAIR	**Ask for the repeated-trial data. If no spread can be examined, a precision claim is unsupported.**

R

Cumulative Review Key

Answers from all lessons

TEACHER KEY

R1. Rewrite '.075kgs' as a correct SI-style record.

ANSWER: 0.075 kg.

R2. Write 0.0008200 in scientific notation and state its significant figures.

ANSWER: 8.200×10^{-4}; four significant figures.

R3. Calculate 6.20 x 4.1 and report the correct significant figures.

ANSWER: 25; the product 25.42 rounds to two significant figures.

R4. Calculate 12.08 + 0.6 + 3.214 and apply the decimal-place rule.

ANSWER: 15.9.

R5. Convert 2.75 h to seconds using dimensional analysis.

ANSWER: 2.75 h x 60 min/h x 60 s/min = 9.90×10^{3} s.

R

Cumulative Review Key

TEACHER KEY

Answers from all lessons

R6. A sample has mass 24.96 g and volume 10.00 mL. Calculate density.

ANSWER: 2.496 g/mL.

R7. A material has density 3.60 g/mL and volume 7.25 mL. Calculate mass.

ANSWER: 26.1 g.

R8. Experimental density is 2.40 g/mL; accepted density is 2.50 g/mL. Calculate percent error.

ANSWER: 4.00%.

R9. Trials 24.9, 25.0, and 25.1 g surround an accepted value of 25.0 g. Evaluate accuracy and precision.

ANSWER: Precise and accurate: trials cluster tightly and center on the accepted value.

R10. A conversion result has the unit kg·mL/g when the target is mL. Diagnose the setup.

ANSWER: An unwanted mass unit did not cancel; reverse or replace the mass conversion factor before calculating.

TEACHER KEY

Unit Test A - Rationale Key

Ten items per page

1. B - The sample mass is 12.40 g.

A quantitative observation includes a measured number and unit: 12.40 g.

2. C - Graduated cylinder

A graduated cylinder is designed for liquid-volume measurement and supports finer readings than a beaker.

3. A - 0.45 g

The record uses a leading zero, a space, and the correct lowercase unit symbol without a plural s.

4. D - 82.5 mL

Multiply liters by the exact factor 1000 mL/L: 0.0825 L = 82.5 mL.

5. B - 5.070 x 10^-4

A normalized coefficient lies between 1 and 10, and the recorded trailing zero preserves four significant figures.

6. A - 3

The decimal point shows that both trailing zeros are recorded as significant, giving three significant figures.

7. C - 39.7

Both factors have three significant figures, so 39.68 rounds to 39.7.

8. D - 22.1

The tenths place is least precise, so the sum 22.126 rounds to 22.1.

9. A - 3600 s / 1 h

Hours must cancel, so hours belong in the denominator and seconds in the numerator.

10. C - 25.0 m/s

90.0 x 1000 / 3600 = 25.0 m/s, retaining three significant figures.

Unit Test A - Rationale Key

TEACHER KEY

Ten items per page

11. B - 2.85 g/mL

Density is mass divided by volume: 57.0 / 20.0 = 2.85 g/mL.

12. D - 31.6 g

m = dV = 1.24 x 25.5 = 31.62 g, which rounds to 31.6 g.

13. C - 3.33%

|12.4 - 12.0| / 12.0 x 100 = 3.33%.

14. A - The trials are precise.

The tight grouping supports precision. Accuracy requires comparison with an accepted value.

15. D - Accuracy

Accuracy describes agreement with an accepted value.

16. B - Read at eye level at the bottom of the meniscus.

Eye-level viewing at the bottom of a concave meniscus reduces parallax error.

17. A - Precision requires repeated trials.

Precision is agreement among repeated measurements, so a single percent-error value is insufficient.

18. D - Reorient a factor so the unwanted unit cancels.

Uncanceled units reveal an incomplete setup; conversion factors must be oriented by unit cancellation.

19. C - Density is intensive.

Density does not depend on sample amount under the same conditions, so it is intensive.

20. B - 2.85 g/mL

The value includes a justified numerical result and the compound unit required for density.

Unit Test B - Rationale Key

B

TEACHER KEY

Ten items per page

1. A - The crystal is colorless.

Colorless is a descriptive observation without a measured number.

2. B - Electronic balance

An electronic balance directly measures mass.

3. D - 6.0 mL

Milliliter uses lowercase m, uppercase L, a space after the value, and no plural s.

4. C - 7.4 mg

1000 micrograms = 1 milligram, so 7400 micrograms = 7.4 mg.

5. C - 812,000

A positive exponent of 5 moves the decimal five places right: 812,000.

6. D - 5

The captive zero and both trailing decimal zeros are significant, for five total.

7. A - 18

The divisor has two significant figures, so 18.24 rounds to 18.

8. B - 99.6

100.0 is precise to the tenths place, so 99.63 rounds to 99.6.

9. D - 86.4 h

3.60 days x 24 h/day = 86.4 h; 24 is an exact defined factor.

10. B - 64.8 km/h

18.0 x 3600 / 1000 = 64.8 km/h.

B

Unit Test B - Rationale Key

TEACHER KEY

Ten items per page

11. C - 2.98 g/mL

Density equals mass divided by volume: 44.7 g / 15.0 mL = 2.98 g/mL.

12. A - 5.96 cm^3

V = m/d = 14.9 / 2.50 = 5.96 cm^3.

13. B - 4.00%

|24.0 - 25.0| / 25.0 x 100 = 4.00%.

14. D - Accurate on average but not precise

The mean supports accuracy, but the wide spread shows low precision.

15. B - Precise but not accurate

Tight grouping shows precision; consistent displacement from the accepted value shows inaccuracy.

16. A - A systematic offset

A constant zero offset shifts measurements in one direction and is systematic.

17. C - They reveal the spread and support a precision judgment.

Repeated trials show variation and therefore provide evidence about precision.

18. A - It is a defined relationship.

Defined unit relationships are exact rather than measured.

19. D - One additional recorded decimal place

The trailing decimal zero records an additional place of precision; it does not change the magnitude.

20. A - 5.96 cm^3

A quantitative result must include its justified value and unit.

